
To Forge or Not to Forge: Predicting Task-Level Fine-Tuning Gains from Ten-Example Pilots

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Abstract

1 Fine-tuning a small specialist model is the default path to cheap task-specific capa-
2 bility, yet the upstream decision — is this task worth fine-tuning at all? — is still
3 made by intuition or by paying for the full training run. We ask whether ten labeled
4 examples can answer it in advance. Under a single frozen protocol (Qwen2.5-
5 1.5B-Instruct, fixed LoRA recipe and budgets, greedy decoding, execution-based
6 verifiers), we measure 61 tasks — 11 anchored in the small-model post-training
7 literature and 50 drawn by a pre-registered filter from Super-NaturalInstructions
8 — pairing a ten-example pilot with a full fine-tune whose measured gain serves as
9 ground truth. All analyses were pre-registered. We find: (1) fine-tuning helps far
10 more often than not — 53 of 61 tasks (87%) show positive full-training gains, mak-
11 ing “forge” the empirical default in this regime; (2) pilot signals significantly rank
12 gain magnitude (Spearman 0.755, 95% CI [0.60, 0.86]), enabling budget alloca-
13 tion by ranking; (3) the pre-registered binary go/no-go endpoint is not established
14 (leave-one-task-out AUC 0.755, bootstrap CI [0.50, 0.93]), a power limit rooted
15 in the empirical scarcity of no-go tasks itself; and (4) description-only features
16 predict below chance (AUC 0.31) — the decision-relevant information exists only
17 in actually running the pilot. We characterize the no-go tail, including negative
18 transfer on heavily post-trained bases (GSM8K -0.23 despite in-domain training),
19 and release the benchmark, protocol, and all sealed artifacts. These sealed mea-
20 surements and pre-registered analyses were produced by an autonomous discover-
21 pilot-decide-train-validate system; without that system’s contribution the 61-task
22 ground-truth table would not exist.

23 **Keywords:** fine-tuning, evaluation, small language models

24 1 Autonomous research result

25 1.1 Setup

26 The operational question is whether a ten-example LoRA probe can tell, in advance, whether full
27 fine-tuning of a small specialist will raise greedy pass@1. We test that question under one frozen
28 protocol, with the analysis contract written down before the remaining production runs, and with an
29 honesty clause that a non-significant AUC is published as a negative result.

30 **Protocol.** The base is Qwen2.5-1.5B-Instruct [Qwen Team, 2024], weights pinned at birth
31 (989aa798...; full sha in the protocol file). LoRA rank 16, $\alpha = 32$, dropout 0.05, on attention
32 and MLP projections; sequence length 4096; per-device batch $2 \times$ accumulation 8 (effective batch
33 16); learning rate 1.0×10^{-4} cosine, warmup 0.03; completion-only loss on the chat template. Pilot:
34 10 examples, 100 steps, seed 20260820. Full: cap 8000, 3 epochs, same seed. Eval: 200-example

slice, seed 20260820, greedy decoding. pass@8 ($k=8$, temperature 1.0, top- p 0.95) is a signal, never the label. Labels are greedy pass@1; Δ is the difference from the unadapted base. go $\Leftrightarrow \Delta_{\text{full}} > 0$ (strict; zeros are no-go).

Sample. The sealed analysis set is $n=61$ tasks: 11 anchored in the small-model post-training literature (Table 2) and 50 drawn from Super-NaturalInstructions [Wang et al., 2022] by a pre-registered English-and-size filter, stratified on Categories[0], seed 20260822, with Source[0] capped at 3. The intended literature layer was 13; BIRD and introductory APPS produced no go-label and are excluded for logistics, before any label existed. The sample contains 53 go and 8 no-go tasks ($53/61 \approx 86.9\%$). Two long-context tasks (TyDiQA, SuperNI task419) entered after a disclosed OOM fallback that keeps effective batch 16 ($\Delta_{\text{full}} + 0.530 / +0.505$). ARC-Easy is in-sample as PARTIAL with metrics present.

Predictors and intervals. The primary model is a task-level leave-one-task-out logistic regression on the frozen numeric signal vector, including a pre-declared format-compliance scalar. Interval estimates are task-level bootstrap 95% CIs on the LOTO pairs (y_i, p_i) . The secondary endpoint is Spearman $\rho(\Delta_{\text{pilot}}, \Delta_{\text{full}})$. A text-only logistic that never sees a probe — train size, max_new_tokens, prompt-style length, eval n — is the no-pilot control. Single-feature logistics are reported alongside. Cluster-level AUCs were pre-declared and are not estimable.

1.2 Results

Four findings, in the frozen abstract order. Every interval below is the sealed task-level bootstrap 95% CI.

Finding 1: forging is the default. Fine-tuning helps far more often than not: 53 of 61 tasks (87%) have $\Delta_{\text{full}} > 0$. Figure 1 is the whole point: the gate $\Delta_{\text{full}} > 0$ is an easy event on this 1.5B protocol. The scarce class is no-go (8 of 61).

Finding 2: pilots rank gain magnitude. Spearman $\rho(\Delta_{\text{pilot}}, \Delta_{\text{full}}) = 0.755$, CI [0.595, 0.864], which excludes 0. Figure 2 is upward-sloping: the probe ranks *how much* full fine-tuning helps. GSM8K and MATH sit in the third quadrant (both deltas negative) and are labeled. Sensitivities (a) drop ARC-Easy and (b) drop EM-floor rows leave Spearman intact: 0.756 [0.594, 0.873] and 0.771 [0.608, 0.878]; both CIs exclude 0.

Finding 3: the binary gate is not established. The main LOTO logistic has AUC 0.755 with CI [0.500, 0.929]. The interval covers 0.5. Under the pre-registered honesty clause this is a negative result: a 10-example LoRA probe does not significantly classify whether full fine-tuning will raise greedy pass@1. The same verdict holds in the two pre-declared sensitivities, reported next to the main row rather than as afterthoughts (Table 1): (a) drop ARC-Easy, $n=60$, AUC 0.748 [0.487, 0.920]; (b) drop EM-floor rows, $n=57$, AUC 0.778 [0.411, 0.985]. Both CIs cover 0.5. Single-feature Δ_{pilot} is weaker (0.663 [0.480, 0.825]); generation length is 0.818 [0.505, 0.986] and covers 0.5 at the edge.

Finding 4: description features run below chance. The text-only control, which never sees a probe or a run metric, has AUC 0.309 [0.045, 0.622] — below chance. Task size and prompt length do not explain go. Decision-relevant information exists only in actually running the pilot.

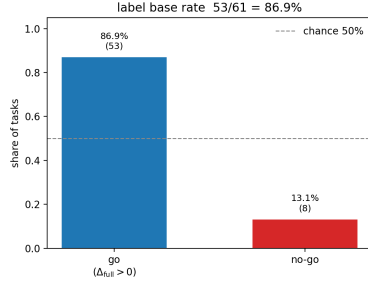


Figure 1: Label base rate $53/61 = 86.9\%$.

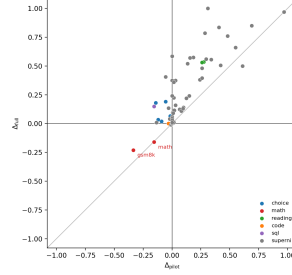


Figure 2: Pilot vs. full gain. GSM8K/MATH labeled in Q3.

Table 1: Primary LOTO AUC and pre-declared sensitivities. Covering 0.5 is a negative under the honesty clause.

Model	n	AUC	95% CI	covers 0.5
Main LOTO logistic	61	0.755	[0.500, 0.929]	yes
(a) drop ARC-Easy	60	0.748	[0.487, 0.920]	yes
(b) drop EM-floor	57	0.778	[0.411, 0.985]	yes
Text-only control	61	0.309	[0.045, 0.622]	yes (below chance)

Table 2 reports the 11 literature-layer tasks that are in the sealed sample. BIRD and APPS are out (logistics exclude, no label).

Table 2: Literature-layer tasks in $n=61$. Display rounding matches the sealed reports; no new analysis.

Task	base	Δ_{pilot}	Δ_{full}
GSM8K	0.690	-0.335	-0.230
WinoGrande	0.560	-0.055	+0.190
Spider	0.495	-0.155	+0.150
MATH	0.355	-0.155	-0.160
MBPP	0.055	-0.030	+0.000
DROP	0.055	+0.275	+0.535
TyDiQA	0.170	+0.260	+0.530
ARC-Easy	0.850	-0.120	+0.035
ARC-Challenge	0.780	-0.090	+0.020
HellaSwag	0.650	-0.140	+0.180
PIQA	0.765	-0.015	+0.065

1.3 Discussion

The upstream decision — is this task worth fine-tuning? — is still made by intuition or by paying for the full run. A cheap, execution-based probe that ranked tasks would change how small-model post-training budgets are allocated. The measurements above show that such a probe exists as a ranker and does not exist as a binary gate, in a regime where forging is already the default.

The honest reading is split. If the operational question is *whether* to run full fine-tuning, the 10-shot probe is not a significant classifier (Finding 3). If the question is *how large* the full-training gain will be, the same probe ranks tasks (Finding 2). Those two sentences should not be collapsed: the probe is a ranker, not a stoplight.

Finding 1 is the mechanism of Finding 3, not a nuisance. With eight negatives, AUC is under-powered for a hard gate. A classifier that always predicts go is already strong in accuracy and useless in ROC. Missing negatives include two logistics exclusions: BIRD never received metrics; APPS never received Δ_{full} . APPS reached base pass@1 0.0 / pass@8 0.005 before the full-training OOM; that floor is a limitations footnote, not a sample row.

90 **Failure modes in the sample.** Format overwrite: GSM8K $\Delta_{\text{full}} = -0.230$ despite in-domain
91 training. The base already emits the dataset’s ##### format; full training saturates format while
92 reasoning degrades. MATH sits in the same quadrant ($\Delta_{\text{pilot}} = -0.155$, $\Delta_{\text{full}} = -0.160$). Exact-
93 match floor: four SuperNI rows have $\Delta_{\text{full}} = 0$ because the matcher never hits (task071, task1553,
94 task236, task455); sensitivity (b) drops them and the negative gate remains. OOM disclosure:
95 TyDiQA and task419 are in-sample only because of the 1×16 fallback (effective batch still 16).
96 WinoGrande, often treated as a weak-gain control in this size class, shows $\Delta_{\text{full}} = +0.190$ here —
97 a reverse relative to that prior, not a new endpoint.

98 **Related work (lite).** Rectified Scaling Law [Lin et al., 2024] predicts pretraining loss versus com-
99 pute in order to choose which model to fine-tune; we instead ask, for a fixed small model, which
100 tasks are worth fine-tuning at all. The Fine-Tuning Trap [Nair and Tao, 2026] documents negative
101 transfer of PEFT on sub-1B mathematical reasoning; our GSM8K/MATH third-quadrant points are
102 a task-level counterpart under a frozen LoRA recipe. Super-NaturalInstructions [Wang et al., 2022]
103 supplies the 50-task slice; CodeS [Li et al., 2024] is the text-to-SQL line that motivates Spider as an
104 anchor.

105 We do not add features, change the go threshold, or reopen BIRD or APPS. Cluster AUCs stay
106 unreported as estimates: the choice cluster is all-go; the math cluster has $n=2$.

107 2 System Design and Meta-Analysis

108 2.1 Agent and Harness

109 [OPERATOR PROSE HERE]

110 2.2 Tools

111 [OPERATOR PROSE HERE]

112 2.3 Research Loop

113 [OPERATOR PROSE HERE]

114 2.4 Human Interventions

115 [OPERATOR PROSE HERE]

116 2.5 Verification

117 [OPERATOR PROSE HERE]

118 2.6 Significance of result

119 [OPERATOR PROSE HERE]

120 2.7 Interpretability

121 [OPERATOR PROSE HERE]

122 2.8 Limitations

123 [OPERATOR PROSE HERE]

124 Material pack

125 Intervention-point checklist.

- 126 1. Protocol freeze at v2 (no v3). Pilot 10×100 kept as the probe.
- 127 2. Sample-frame correction: pool rows $\neq$ tasks; intended 13 literature +50 SuperNI = 63.
- 128 3. Dual-worker plan cancelled; one serial queue of 60.
- 129 4. BIRD packed as a separate later package; Mini-Dev never used.
- 130 5. False-negative remote-shell diagnosis (tool layer, not host); production started after the
- 131 correction.
- 132 6. APPS wall: 3h then 8h ops fence, declared before any APPS label.
- 133 7. BIRD checker aligned to dual zip-sha fields (not a single sha256 key).
- 134 8. Failed run directories isolated, not deleted.
- 135 9. OOM fallback 1×16 only for TyDiQA and SuperNI task419; effective batch unchanged.
- 136 10. Campaign close: $n=61$; BIRD/APPS logistics exclude before any label.

137 System-number archive.

- 138 • Sealed sample $n=61$ (53 go / 8 no-go).
- 139 • Serial queue 60/60 completed; two literature-layer tasks produced no label.
- 140 • Protocol yaml never edited after freeze.

- Disclosed in-sample exceptions: ARC-Easy PARTIAL; TyDiQA/task419 OOM fallback ($\Delta_{\text{full}} + 0.530 / +0.505$).
- APPS S1 base pass@1 0.0 / pass@8 0.005: limitations footnote, not a row.
- Total GPU-hours: not computed; do not invent.

Architecture (material figure). Figure 3 is a schematic of the factory–queue–gates–harvest loop used to produce the sealed table. Caption is factual, not Part 2 narrative.

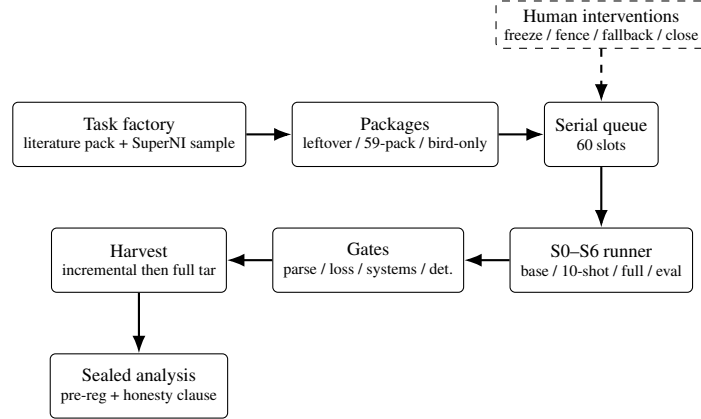


Figure 3: Factory → serial queue → per-task S0–S6 runner → gates → harvest → sealed analysis. Dashed boxes: human intervention points.

147 **3 Broader Impact**

148 [OPERATOR PROSE HERE]

149 **Reflection-candidate list**

- 150 1. Tool-layer false-negative connectivity vs. actual host failure: a process risk for autonomous
151 long jobs.
- 152 2. Schema lag (checker vs. dual zip-sha fields) vs. logistics: labels must not be invented to
153 close a queue.
- 154 3. Format overwrite vs. EOS as the GSM8K mechanism ($\Delta_{\text{full}} = -0.230$).
- 155 4. Wall-clock as ops fence vs. measurement semantics (APPS 3h then 8h, still no Δ_{full}).
- 156 5. Negative-class scarcity (8 no-go / 61) as a first-class design constraint for any go/no-go
157 endpoint.
- 158 6. Description-only features running below chance (AUC 0.309): skipping the GPU is not a
159 cheap substitute.
- 160 7. How the field should adapt: pre-register the honesty clause before the remaining runs;
161 freeze the protocol; do not reopen logistics excludes after seeing labels.

A Reproducibility Statement

The protocol file, pre-registration, and the sealed 61-row per-task table are packed in the supplementary archive. Protocol hyperparameters were never edited after freeze. Dataset papers for the literature-layer table: Disclosed run-level exceptions (OOM fallback on two tasks; one PARTIAL in-sample row) are listed in Setup and in the supplementary protocol notes. Figures in Part 1 are redrawn from the frozen plot payload; they are not a new analysis.

B Disclosure Statement

Agent(s) / model(s) used

- ☐ A human author has curated this work and verified every claim, figure, methodology detail, and reference.
- ☐ A human author takes full accountability for the submission as a human-authored, human-verified artifact.

References

- Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*, 2021.
- Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, and Yejin Choi. PIQA: Reasoning about physical commonsense in natural language. *arXiv preprint arXiv:1911.11641*, 2019.
- Jonathan H. Clark, Eunsol Choi, Michael Collins, Dan Garrette, Tom Kwiatkowski, Vitaly Nikolaev, and Jennimaria Palomaki. TyDi QA: A benchmark for information-seeking question answering in typologically diverse languages. *arXiv preprint arXiv:2003.05002*, 2020.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- Dheeru Dua, Yizhong Wang, Pradeep Dasigi, Gabriel Stanovsky, Sameer Singh, and Matt Gardner. DROP: A reading comprehension benchmark requiring discrete reasoning over paragraphs. *arXiv preprint arXiv:1903.00161*, 2019.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. *arXiv preprint arXiv:2103.03874*, 2021.
- Haoyang Li, Jing Zhang, Hanbing Liu, Ju Fan, Xiaokang Zhang, Jun Zhu, Renjie Wei, Hongyan Pan, Cuiping Li, and Hong Chen. CodeS: Towards building open-source language models for text-to-SQL. *arXiv preprint arXiv:2402.16347*, 2024.
- Haowei Lin, Baizhou Huang, Haotian Ye, Qinyu Chen, Zihao Wang, Sujian Li, Jianzhu Ma, Xiaojun Wan, James Zou, and Yitao Liang. Selecting large language model to fine-tune via rectified scaling law. *arXiv preprint arXiv:2402.02314*, 2024.
- Rahul Nair and Chun Tao. The fine-tuning trap: Evaluating negative transfer and the role of PEFT in sub-1B mathematical reasoning. *arXiv preprint arXiv:2606.06920*, 2026.
- Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. WinoGrande: An adversarial winograd schema challenge at scale. *arXiv preprint arXiv:1907.10641*, 2019.

- 207 Yizhong Wang, Swaroop Mishra, Pegah Alipoormolabashi, Yeganeh Kordi, Amirreza Mirzaei, et al.
208 Super-naturalinstructions: Generalization via declarative instructions on 1600+ NLP tasks. *arXiv*
209 *preprint arXiv:2204.07705*, 2022.
- 210 Tao Yu, Rui Zhang, Kai Yang, Michihiro Yasunaga, Dongxu Wang, Zifan Li, James Ma, Irene Li,
211 Qingning Yao, Shanelle Roman, Zilin Zhang, and Dragomir Radev. Spider: A large-scale human-
212 labeled dataset for complex and cross-domain semantic parsing and text-to-SQL task. *arXiv*
213 *preprint arXiv:1809.08887*, 2018.
- 214 Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. HellaSwag: Can a
215 machine really finish your sentence? *arXiv preprint arXiv:1905.07830*, 2019.